D D M M Y Y Y Y

Assignment - II

01] Let $A = \{1,2,3,4,6\}$ and R be a relation on A defined by aRb if and only if "a is a multiple of b". Write down the relation R, relation matrix M(R). And draw its diagraph. List out its in degree and out degree.

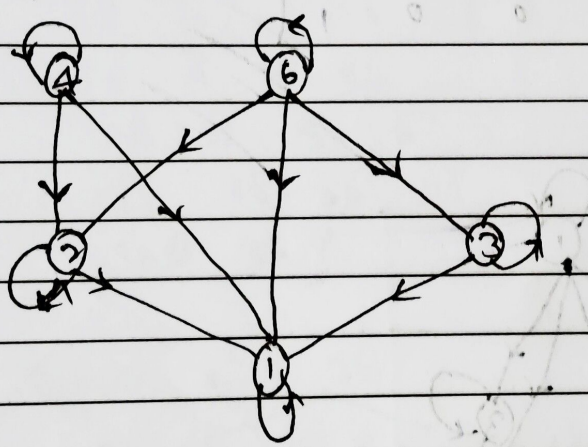
→ $R = \{(1,1), (2,1), (2,2), (3,1), (3,3), (4,1), (4,2), (4,4), (6,1), (6,2), (6,3), (6,6), (6,6)\}$.

Relational matrix R M(R).

$M(R) =$

	1	2	3	4	6
1	1	0	0	0	0
2	1	1	0	0	0
3	1	0	1	0	0
4	1	1	0	1	0
6	1	1	1	0	1

Diagraph



Indegree and outdegree

vertices	In-degree	out-degree
1	5	1
2	3	2
3	2	2

D D M M Y Y Y Y

4 1 3
6 1 4

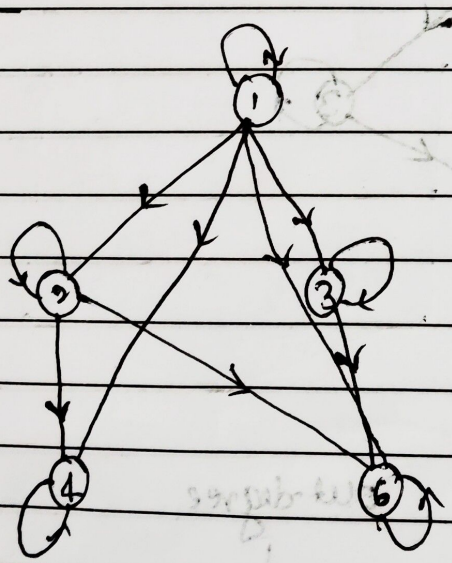
02] Let $A = \{1, 2, 3, 4, 6\}$ and R be a relation on A defined by aRb if and only if "a divides b". Write down the relation R, relation matrix M(R). And draw it's diagraph also find its indegree and out degree.

→ $R = \{(1,1), (1,2), (1,3), (1,4), (1,6), (2,2), (2,4), (2,6), (3,3), (3,6), (4,4), (6,6)\}$

Relational matrix M(R)

M(R) =	1	2	3	4	6
1	1	1	1	1	1
2	0	1	0	1	1
3	0	0	1	0	1
4	0	0	0	1	0
6	0	0	0	0	1

Diagraph



D D M M Y Y Y Y

Indegree & outdegree

vertices	Indegree	outdegree
1	1	5
2	2	3
3	2	2
4	3	1
6	4	1

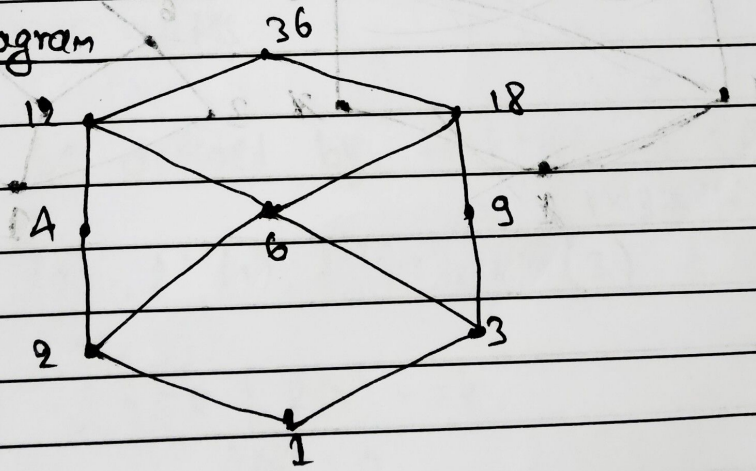
03] Draw the Hasse Diagram which represents positive divisors of 36.

→ The positive divisers of 36

$A = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$

$R = \{$ 1R 1, 2, 3, 4, 6, 9, 12, 18, 36
2R 2, 4, 6, 12, 18, 36
3R 3, 6, 9, 12, 18, 36
4R 4, 12, 36, 6R 6, 12, 18, 36
9R 9, 18, 36
12R 12, 36
18R 18, 36
36R 36 $\}$

Hasse diagram



D D M M Y Y Y Y

Q4] Let $A = \{1, 2, 3, 4, 6, 8, 12\}$ and define R on A by xRy iff 'x divides y'. Prove that (A, R) is POSET. Draw the Hasse diagram for (A, R)

→ $R = \{(1,1), (1,2), (1,3), (1,4), (1,6), (1,8), (1,12)$
$(2,2), (2,4), (2,6), (2,8), (2,12), (3,3), (3,6), (3,12)$
$(4,4), (4,8), (4,12)$
$(6,6), (6,12)$
$(8,8)$
$(12,12)\}$

T.P.T R is POSET

$(1,1), (2,2), (4,4), (3,3), (6,6), (8,8), (12,12) \in R$ ∴ R is Reflexive

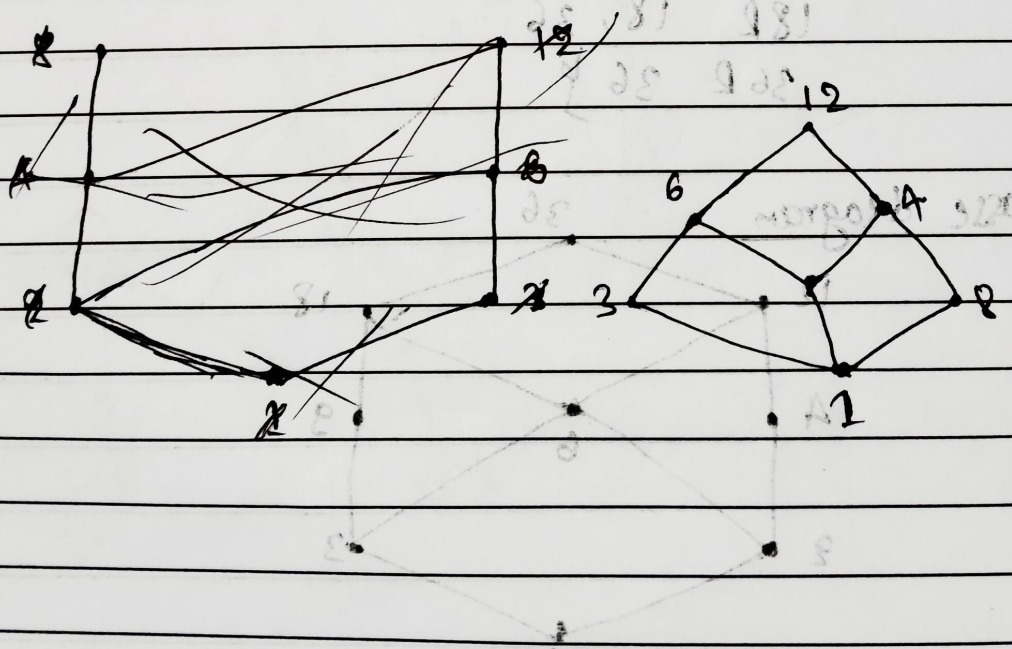
$(1,2) \in R$ but $(2,1) \notin R$ ∴ R is asymmetric

$\forall a,b,c \in A$ $(a,b) \in R, (b,c) \in R$ then $(a,c) \in R$

∴ R is a transitive

∴ R is a POSET

Hasse diagram



05) If $f: A \to B$, $g: B \to C$, $h: C \to D$ are three functions, then prove that $ho(gof) = (hog)of$

→ LHS $= ho\{g(of)\}$

$= ho g\{f(x)\}$

let $f(x) = y$

$= ho g\{y\}$

let $g(y) = z$

$= ho\{g(y)\}$

$= ho\{z\}$

$= h(z)$

$(hog)of = h(z)$ ——①

LHS $= ho(gof)$

$= h\{gof\}$

$= h\{g\{f(x)\}\}$

let $f(x) = y$

$= h\{g(y)\}$

let $g(y) = z$

$= h\{z\}$

$ho(gof) = h(z)$ ——②

FROM ① & ②

LHS = RHS

06) Let $f: R$ be defined by, $f(x) = \begin{cases} 3x-5 & \text{if } x > 0 \\ 1-3x & \text{if } x \le 0 \end{cases}$

Find $f^{-1}(0)$, $f^{-1}(1)$, $f^{-1}(-1)$, $f^{-1}(3)$, $f^{-1}(6)$, $f^{-1}([-6,6])$ and $f^{-1}([-5,5])$

→ i) $f^{-1}(0) = \{x \in R \mid f(x) = 0\}$

$f(x) = 0$ $\quad f(x) > 0$ $\quad f(x) \le 0$

D D M M Y Y Y Y

$f(x) \geq 0$	$f(x) > 0$	$f(x) < 0$
$f(x) = 1-3x$	$f(x) = 3x-5$	$f(x) = 1-3x$
$0 = 1-3x$	$0 = 3x-5$	$0 = 1-3x$
$-1 = -3x$	$5 = 3x$	$-1 = -3x$
$x = 1/3$	$x = 5/3$	$x = 1/3$
($1/3 \geq 0$ not possible)	(possible)	(not possible)

$f^{-1}(0) = \{5/3\}$

(ii) $f^{-1}(1)$ $x > 0$	$f^{-1}(1)$ $x \leq 0$
$f(x) = 3x-5$	$f(x) = 1-3x$
$1 = 3x-5$	$1 = 1-3x$
$6 = 3x$	$0 = -3x$
$x = 2$	$x = 0$
(possible value)	(possible value)

$f^{-1}(1) = \{2, 0\}$

(iii) $f^{-1}(-1)$ $x > 0$	$f^{-1}(-1)$ $x \leq 0$
$f(x) = 3x-5$	$f(x) = 1-3x$
$-1 = 3x-5$	$-1 = 1-3x$
$4 = 3x$	$-2 = -3x$
$x = 4/3$	$x = 2/3$
(possible value)	(not possible value)

$f^{-1}(-1) = \{4/3\}$

iv) $f^{-1}(3)$ $x>0$

$f(x) = 3x-5$

$3 = 3x-5$

$8 = 3x$

$x = 8/3$

(possible value)

$f^{-1}(3)$ $x \leq 0$

$f(x) = 1-3x$

$3 = 1-3x$

$-2 = -3x$

$x = -2/3$

(possible value)

$f^{-1}(3) = \{8/3, -2/3\}$

v) $f^{-1}(6)$ $x>0$

$f(x) = 3x-5$

$6 = 3x-5$

$11 = 3x$

$x = 11/3$

(possible value)

$f^{-1}(6)$ $x \leq 0$

$f(x) = 1-3x$

$6 = 1-3x$

$5 = -3x$

$x = -5/3$

(possible value)

$f^{-1}(6) = \{11/3, -5/3\}$

vi) $f^{-1}([-6,5]) = \{x \in \mathbb{R} \mid f(x) \in [-6,5]\}$

$= \{x \in \mathbb{R} \mid -6 \leq f(x) \leq 5\}$

$x \geq 0$

$f(x) = 3x-5$

We have $-6 \leq f(x) \leq 5$

add $+6$ throughout

$0 \leq 3x+1 \leq 11$

$-6 \leq 3x-5 \leq 5$

$0 \leq 3x+1 \leq 11$

divide

$x > 0$

$f(x) = 3x-5$

$-6 \leq f(x) \leq 5$

$-6 \leq 3x-5 \leq 5$

add $+5$ throughout

$-1 \leq 3x \leq 10$

$0 \leq 3x+1 \leq 11$

$-1 \leq 3x \leq 10$

divide by 3

$-1/3 \leq x \leq 10/3$

D D M M Y Y Y Y

~~divide by 3~~

But $y > 0$ so:

$0 \leq y \leq 10/3$

$x \leq 0$

$f(x) = 1 - 3x$

$-6 \leq 1 - 3x \leq 5$

sub -1 throughout

$-7 \leq -3x \leq 4$

devide -3 throughout

$7/3 \leq x \leq -4/3$

$x \leq 0$

$0 \leq x \leq -4/3$

$f^{-1}[(-6, 5)] = \{10/3, -4/3\}$

vii) $f^{-1}[(-5, 5)] = \{x \in R \mid f(x) \in (-5, 5)\}$

$= \{x \in R \mid -5 \leq f(x) \leq 5\}$

$x > 0$	$x \leq 0$
$f(x) = 3x - 5$	$f(x) = 1 - 3x$
we have $-5 \leq f(x) \leq 5$	we have $-5 \leq 1 - 3x \leq 5$
$-5 \leq 3x - 5 \leq 5$	$-5 \leq 1 - 3x \leq 5$
~~devi~~ add $+5$	sub -1
$0 \leq 3x \leq 10$	$-6 \leq -3x \leq 4$
divide 3	devide -3
$0 \leq x \leq 10/3$	$2 \leq x \leq -4/3$

$f^{-1}[(-5, 5)] = \{10/3, -4/3\}$

07 Let $A = \{1, 2, 3, 4, 5\}$. Define a relation R on $A \times A$ by (x_1, y_1) R (x_2, y_2) iff $(x_1 + y_1 = x_2 + y_2)$

i) Verify that R is an equivalence relation.

ii) Determine the equivalence class of $[(2,4)]$.

→ i) R is a equivalence relation.

T.P.T R is Reflexive, symmetric, transitive

$\forall x, y \in A \times A$

we have $(x, y) \in (x, y)$ iff $x + y = x + y$

$\therefore (x, y) \in (x, y)$

$\therefore$ R is Reflexive.

R is symmetric

$\forall (x_1, y_1)$ and $(x_2, y_2) \in A \times A$

$(x_1, y_1)\ R\ (x_2, y_2) \Rightarrow x_1 + y_1 = x_2 + y_2$

$= x_2 + y_2 = x_1 + y_1$

$= (x_2, y_2)\ R\ (x_1, y_1)$

$\therefore$ R is a symmetric.

R is transitive

$\forall (x_1, y_1), (x_2, y_2)$ and $(x_3, y_3) \in A \times A$

we have

$(x_1, y_1)\ R\ (x_2, y_2)$ and $(x_2, y_2)\ R\ (x_3, y_3)$

$x_1 + y_1 = x_2 + y_2$ and $x_2 + y_2 = x_3 + y_3$

(substitute $x_2 + y_2 = x_3 + y_3$)

$x_1 + y_1 = x_3 + y_3$

$\therefore$ R is transitive.

$\therefore$ R is Equivalence relation.

11) $[2,4] = \{(x,y) \in A \times A \text{ s.t. } (x,y) R (2,4)\}$

$(x+y=6)$

$= \{(1,5), (2,4), (3,3), (4,2), (5,1)\}$.

8) Given $A=B=C=R$ and functions $f: A \to B$, $f(a) = 2a+1$ and $g: B \to C$, $g(b) = 1/3\, b$, for any $a \in A$ and $b \in B$. Compute gof. Show that gof is invertible. Find $(gof)^{-1}$.

→ $f(a) = 2a+1$ $\quad g(b) = 1/3\, b$

gof

$g\{f(x)$

$g\{f(a)\}$

$g\{2a+1\}$

$g(b) = 1/3\, b$

put $b = 2a+1$

$g(2a+1) = \frac{2a+1}{3}$

$\Rightarrow \frac{2a+1}{3}$.

$g(2a+1) = \frac{2a+1}{3}$

assume

$g \quad \frac{2a_1+1}{3} = \frac{2a_2+1}{3}$

divide 3 on both side

$2a_1+1 = 2a_2+1$

$2a_1 = 2a_2$

$a_1 = a_2$

∴ one-one function.

∴ It is invertable.

TO find $(gof)^{-1}$,

Let $y = \frac{2b+1}{3}$

$3y = 2b+1$

$3y - 1 = 2b$

$b = \frac{3y-1}{2}$

$\Rightarrow (gof)^{-1} = \frac{3y-1}{2}$

09] State pigeon hole principle. Prove that in any set of 29 persons, at least 5 persons have been born on the same day of the week.

→ Statement:

If M pigeon occupies n pigeon holes where $M > n$ then atleast 1 pigeon holes must contain 2 or more pigeons in it.

$n = 7 \quad M = 29$

$$p+1 = \left\lfloor \frac{M-1}{n} \right\rfloor + 1$$

$$p+1 = \left\lfloor \frac{29-1}{7} \right\rfloor + 1$$

$$= \left\lfloor \frac{28}{7} \right\rfloor + 1$$

$$= 4 + 1$$

$$p + 1 = 5$$

∴ Atleast 5 persons have been born on the same day of the week.

11] In how many ways the 26 letters of English alphabet are permuted so that none of the patterns CAR, DOG, PUN or BYTE occurs?

→ let A = AR B = DOG C = PUN D = BYTE

To find $|\bar{A} \cap \bar{B} \cap \bar{C}|$

$|S| = 26!$

$|A| = 24!$ CAR $26-3+1 = 24!$

$|B| = 24!$ DOG $26-3+1 = 24!$

$|C| = 24!$ PUN $26-3+1 = 24!$

$|D| = 23!$ BYTE $26-4+1 = 23!$

$|A \cap B| = 22!$ CAR DOG $26-6+2 = 22!$

$|A \cap C| = 22!$

$|A \cap D| = 22!$

$|B \cap C| = 22!$

$|B \cap D| = 21!$

$|C \cap D| = 21!$

$|A \cap B \cap C \cap D| = 17!$ CAR DOG PUN BYTE $26-13+4 = 17!$

$|A \cap B \cap C| = 20!$

$|B \cap C \cap D| = 19!$

$|C \cap D \cap A| = 19!$

$|D \cap A \cap B| = 19!$

W.K.T

$$|\bar{A} \cap \bar{B} \cap \bar{C} \cap \bar{D}| = |S| - \{|A|+|B|+|C|+|D| - (|A \cap B|+|B \cap C|+|C \cap D|+|A \cap D|+|A \cap C|+|A \cap D|+|B \cap C|+|B \cap D| + (|A \cap B \cap C|+|B \cap C \cap D|+|C \cap D \cap A|+|D \cap A \cap B|) - |A \cap B \cap C \cap D|\}$$

$$|\bar{A} \cap \bar{B} \cap \bar{C} \cap \bar{D}| = 26! - \{(24!+24!+24!+24!) - (22!+22!+21!+22!+21!+21!) + (20!+19!+19!+19!) - 17!\}$$

$|A \cap B \cap C \cap D| = 4.014077864 \times 10^{6}$

v) In how many ways 5 number of a's, 4 no. of b's & 3 no. of c's can be arranged so that all the identical letters are not a single block?

→ To find $|A \cap B \cap C|$

$|A \cap B \cap C| = |S| - \{(|A|+|B|+|C|) - (|A \cap B|+|B \cap C|+|A \cap C|) + |A \cap B \cap C|\}$

$|S| = \frac{12!}{5! \times 4! \times 3!} = 27,720$

$a \to 5 \quad b \to 4 \quad c \to 3$

Total = 12

$|A| = \frac{8!}{1! \times 4! \times 3!} = 280$ — aaaaa (1) bbbb (4) ccc (3) $= \frac{8!}{1! \times 4! \times 3!}$

$|B| = \frac{9!}{5! \times 3! \times 1!} = 504$, $|C| = \frac{10!}{5! \times 4! \times 1!} = 1260$

$|A \cap B| = \frac{7!}{1! \times 1! \times 3!} = 20$ — aaaaa (1) bbbb (1) ccc (3) $= \frac{5!}{1! \times 1! \times 3!}$

$|B \cap C| = \frac{7!}{5! \times 1! \times 1!} = 42$ $|A \cap C| = \frac{6!}{1! \times 4! \times 1!} = 30$

$|A \cap B \cap C| = \frac{3!}{1! \times 1! \times 1!} = 6$ — aaaaa (1) bbbb (1) ccc (1) $= \frac{3!}{1! \times 1! \times 1!}$

put in formula

$|A \cap B \cap C| = 27720 - \{(280+504+1260) - (20+42+30) + 6\}$

$|A \cap B \cap C| = 25,762$

13) The no. of virus affected files in system is 1000 (to start with) and this increases 250% every two hours. Use a recurrence relation to determine the no. of virus affected files in the system after one day.

→ Let a_M be the no. of affected files after n intervals of 2 hours.

$a_0 = 1000$

Increases 250% every 2 hours.

$$a_{M+1} = a_M + 250\%$$
$$= a_M + 2.5$$
$$a_{M+1} = 3.5 a_M$$

After one day,

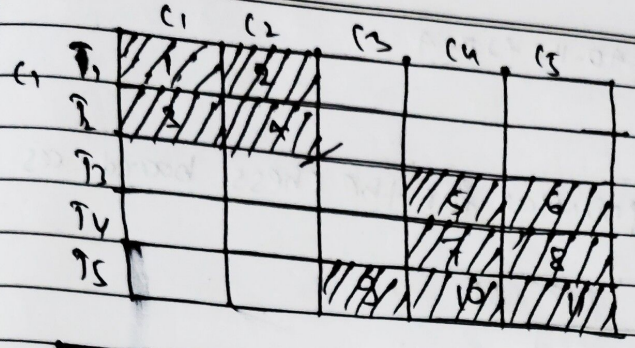
1 day = 24 hrs

$$M = \frac{24}{2} = 12$$

$$a_M = 1000 \times (3.5)^M$$
$$a_{12} = 1000 \times (3.5)^{12}$$
$$a_{12} = 3379,219,600$$

$\langle a_{12} = 3.38 \times 10^9$ files $\rangle$

14) Five teachers T_1, T_2, T_3, T_4, T_5 are to be made class teachers for five classes, C_1, C_2, C_3, C_4, C_5 one teacher for each class. T_1 & T_2 do not wish to become the class teacher for C_1 or C_2, T_3 and T_4 for C_4 or C_5, and T_5 for C_3 or C_4 or C_5. In how many ways can the teachers be assigned the work (without displeasing any teacher).



1	2
3	4

C1

	5	6
	7	8
9	10	11

C2

$r_1 = 0 = 4$

$r_2 = (1,4)\ (2,3) = 2$

$r_3 = 0$

$r(C_1,x) = 1 + 4x + 2x^2 + 0$

$r_1 = 0 = 7$

$r_2 = (5,8), (5,11), (5,9), (6,7), (6,10), (6,9), (7,11), (7,9), (8,10), (8,9) = 10$

$r_3 = (5,8,10), (6,7,11) = 2 \quad r_4 = 0$

$r(C_2,x) = 1 + 7x + 10x^2 + 2x^3 + 0$

$r(C,x) = r(C_1,x) \cdot r(C_2,x)$

$= (1 + 4x + 2x^2) \cdot (1 + 7x + 10x^2 + 2x^3)$

$= 1 + 7x + 10x^2 + 2x^3 + 4x + 28x + 40x^3 + 8x^4 + 2x^2 + 14x^3 + 20x^4 + 4x^5$

$\therefore r = 1 + 11x + 40x^2 + 56x^3 + 28x^4 + 4x^5$

$\therefore r_1 = 11,\ r_2 = 40,\ r_3 = 56,\ r_4 = 28,\ r_5 = 4$

$N = S_0 - S_1 + S_2 - S_3 + S_4 - S_5$

$S_0 = n! = 5! = 120$

$S_1 = (n-1)! \times r_1 = 264$

$S_2 = (n-2)! \times r_2 = 240$

$S_3 = (n-3)! \times r_3 = 112$

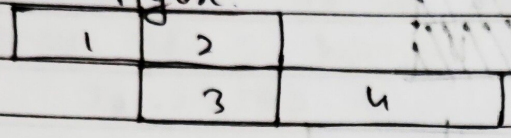
$S_4 = (n-4)! \times r_4 = 28$

$S_5 = (n-5)! \times r_5 = 4$

$\bar{N} = 120 - 264 + 240 - 112 + 28 - 4$

$\bar{N} = 8$

15] Find the rook polynomial for the chess board as shown on the figure



→

$r_1 = 0 = 4$

$r_2 = (1,3), (1,4), (2,4) = 3$

$r_3 = 0$

$r(1,x) = 1 + 4x + 3x^2$

16] Define group. Show that 4th roots of unity is an abelian group.

→ A non empty set $(G, *)$ is said to be a group under the operation $*$ if it follows the following conditions

i) Closure law: $\forall a, b \in G \Rightarrow a * b \in G$

ii) Associative law: $\forall a, b, c \in G \Rightarrow (a*b)*c = a*(b*c)$

iii) Identity law: $\forall a \in G$ & $e \in G$ s.t $a*e = e*a = a$.

iv) Inverse law: $\forall a \in G$ & $a' \in G$ s.t $a*a' = a'*a = e$.

The 4th root of unity is

$W4 = \{1, -1, i, -i\}$

x	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

i) Closure law

$\forall a, b \in W_4 \Rightarrow a*b \in W_4$

$a = 1 \quad b = -1 \rightarrow a \times b$

$\Rightarrow 1 \times -1$

$\Rightarrow -1 \in W_4$

$\therefore$ Closure law satisfied.

ii) Associative law

$\forall a, b, c \in W_4 \Rightarrow a*(b*c) = (a*b)*c$

$a = 1 \quad b = -1 \quad c = i$: $a \times (b \times c) = 1 \times ((-1) \times i) = -1 \times i = -i$

$(a \times b) \times c = (1 \times -1) \times i = -1 \times i = -i$

$a \times (b \times c) = (a \times b) \times c$

Associative law satisfied.

iii) Identity law

$\forall a \in W_4$ & $e \in W_4$

s.t $a*e = a$ & $e*a = a$

$1 \times 1 = 1$

$-1 \times -1 = -1$

$i \times 1 = i$

$e = 1$

Identity law satisfied.

iv) Inverse law

$\forall a \in W_4$ & $a^{-1} \in W_4$ s.t $a*a^{-1} = e$

$a * a^{-1} = e$

$1 \times (1) = 1 \Rightarrow 1^{-1} = 1$

$-1 \times (-1) = 1 \Rightarrow -1^{-1} = 1$

$i \times (-i) = 1 \Rightarrow -i^{-1} = 1$

$-i \times (i) = 1 \Rightarrow i^{-1} = 1$

Inverse law satisfied.

v) COMMUTATIVE law

$\forall a, b \in WA \Rightarrow a * b = b * a$

$a = 1 \quad b = -1 \quad a \times b = -1$

$b \times a = -1$

$a \times b = b \times a$

COMMUTATIVE law is satisfied

$\therefore (WA, \times)$ is an abelian group

17] Define klein group. verify $A = \{1, 3, 5, 7\}$ is a klein 4 group.

→ The group G is said to be klein 4 group if G is abelian and each element of G has self inverse.

$A = \{1, 3, 5, 7\}$

*	1	3	5	7
e=1	(1)	3	5	7
3	3	(1)	7	5
5	5	7	(1)	3
7	7	5	3	(1)

i) closure law

$\forall a, b \in A = a * b \in A$

$1, 3 \in A \quad 1 * 3 \in A$

$\therefore$ closure law satisfied.

ii) Associative law

$\forall a, b, c \in A \quad a * (b * c) = (a * b) * c \in A$

$a = 1 \quad b = 3 \quad c = 5$

$1 * (3 * 5) = (1 * 3) * 5$

$1 * 7 = 3 * 5$

$7 = 7$

Associative law satisfied.

iii) Identity law

$\forall a \in A$ & $e \in A$ s.t $e*a = a$

$a*e = a$

$a = 1$

$1*e = 1$

$1*1 = 1$

$1 = 1$

~~Inverse~~ law satisfied.

Identity

iv) Inverse law

$\forall a \in G$ & $a^{-1} \in A$ s.t $a*a^{-1} = a^{-1}*a$

$1^{-1} = 1 \quad 3^{-1} = 3 \quad 5^{-1} = 5 \quad 7^{-1} = 7$

∴ Inverse law satisfied.

v) Commutative law

$\forall a, b \in A \quad a*b = b*a$

$3*5 = 5*3$

$7 = 7$

$a*b = b*a$

Commutative law satisfied.

vi) Self Inverse

The odd elements have self Inverse.

∴ A is a field & group.

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18] If G be a group with subgroup H and K. If $|G|=660$ and $|K|=66$ & $K \subset H \subset G$ & find the possible value for $|H|$.

→ By Lagrange's theorem,
$|H|$ Must divide $|G|=660$ and
$|K|=66$ must divide $|H|$
Also we have $|G| \neq |H| \neq |K|$
Therefore $660 = |H| q_1$ for some integer $q_1 > 1$ and $|H| = 66q_2$ for some integer $q_2 > 1$
Hence,
$660 = 66 q_1 q_2$ (or) $q_1 q_2 = 10$
Thus either,
$q_1 = 2$ & $q_2 = 5$ (or) $q_1 = 5$ & $q_2 = 2$
$|H| = 66 \times 5 = 330$ (or) $|H| = 66 \times 2 = 132$

19] State and prove Lagrange's Theorem.

→ Statement

→ If G is a finite group & H is a subgroup of G then the $|H|$ divides the $|G|$.

$$\frac{O(G)}{O(H)}$$

Proof

→ Since G is a finite group then H is also finite.
→ Therefore the no of cosets of H in G are also finite.
→ Let $Ha_1, Ha_2 \ldots Ha_r$ be the distinct right cosets of H in G.
→ Then S right cosets decomposition of G.

$$G = Ha_1 \cup Ha_2 \cup \ldots \cup Ha_r$$
$$O(G) = O(Ha_1) + O(Ha_2) + \ldots + O(Ha_r)$$

But,
$$O(Ha_1) = O(Ha_2) \ldots O(Ha_r) = O(H)$$

$o(G) = o(H) + o(H) + \cdots + o(H)$ (r times)

$o(G) = r\, o(H)$

$\frac{o(G)}{o(H)} = r$

$\therefore$ order of H divides order of G.

20] If H, K are subgroups of a group G, prove that $H \cap K$ is also a subgroup of G. Is $H \cup K$ a subgroup of G.

→ Let G be a group & H, K are subgroups of G.

Take any $a\ b \in H \cap K$. Then $a, b \in H$ & $a, b \in K$.

$\therefore ab^{-1} \in H$ & $ab^{-1} \in K$ ($\because$ If $H \leq G$, $a, b \in H \Rightarrow a, b \in H$ & $a^{-1} \in H$)

$\Rightarrow ab^{-1} \in H \cap K$ ($\because$ If $H, K \leq G$ & $a, b \in H, K \Rightarrow ab^{-1} \in H, K$)

Hence $H \cap K$ is a subgroup of G

Now consider the symmetric group S_3 of 2 of it's subgroups $H = \langle P_0, P_3 \rangle$ & $K = \langle P_0, P_4 \rangle$, Then

$H \cup K = \langle P_0, P_3, P_4 \rangle$

From Multiplication table for S_3, we find that $P_3 P_4 = P_1$.

But $P_1 \notin H \cup K$.

$\therefore H \cup K$ is not closed under product of permutations of consequently cannot be a group.

$\therefore H \cup K$ is not a group.